局部拟有限 SYNTOMIC 态射的微分形式比较映射

Counting scope. The primary problem is Tag 0FLA, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题构造对所有局部拟有限 syntomic 态射自然的微分形式到 det(NL) 扭曲微分形式的映射。允许局部模型及基变换引理后，仍须用纤维积比较两个 nice 模型，证明所得局部映射与模型选择无关并能粘合。核心局部构造有明确前置输入，而多模型一致性与全局化是本题承担的实质任务，判 H2。.

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1. Comparing sheaves of differential forms

The goal of this section is to construct for every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ maps

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for all $p \geq 0$ satisfying the following properties

(1) $c^0_{Y/X}(1) = \delta(NL_{Y/X})$ see Discriminants, Section section tate map (Tag 0FKB),
(2) $c^{q+p}_{Y/X}(\omega \wedge \eta) = \omega \wedge c^p_{Y/X}(\eta)$ for local sections ω of $f^*\Omega^q_{X/\mathbf{Z}}$ and η of $\Omega^p_{Y/\mathbf{Z}}$,
(3) the formation of $c^p_{Y/X}$ is compatible with restriction to opens and with base change (as formulated in Lemma 1.2).

Note that these conditions imply that the maps $c^p_{Y/X}$ are $\mathcal{O}_Y$-linear and that the composition with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$. We will in fact show that there exists a unique collection of operators $c^p_{Y/X}$ with properties (1), (2), (3). This result will be applied in Section section trace (Tag 0FK6) to the construction of the trace map on the de Rham complexes if f is finite.

Remark 1.1. Let A be a ring. Let $P = A[x_1, \ldots, x_n]$. Let $f_1, \ldots, f_n \in P$ and set $B = P/(f_1, \ldots, f_n)$. Assume $A \to B$ is quasi-finite. Then B is a relative global complete intersection over A (Algebra, Definition 1.5) and $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the classes $\overline{f}_i$ by Algebra, Lemma lemma relative global complete intersection conormal (Tag 00SV). Consider the following diagram

$$\begin{array}{ccccc} \Omega_{A/\mathbf{Z}} \otimes_A B & \longrightarrow & \Omega_{P/\mathbf{Z}} \otimes_P B & \longrightarrow & \Omega_{P/A} \otimes_P B \\ & & \uparrow & & \uparrow \\ & & (f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 & = & (f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \end{array}$$

The right column represents $NL_{B/A}$ in $D(B)$ hence has cohomology $\Omega_{B/A}$ in degree 0. The top row is the split short exact sequence $0 \to \Omega_{A/\mathbf{Z}} \otimes_A B \to \Omega_{P/\mathbf{Z}} \otimes_P B \to \Omega_{P/A} \otimes_P B \to 0$. The middle

column has cohomology $\Omega_{B/\mathbf{Z}}$ in degree 0 by Algebra, Lemma lemma differential seq (Tag 00RU). Thus by Lemma lemma funny map (Tag 0FL8) we obtain canonical B-module maps

$$\Omega^p_{B/\mathbf{Z}} \longrightarrow \Omega^p_{A/\mathbf{Z}} \otimes_A \det(NL_{B/A})$$

for $p \geq 0$ satisfying properties (1) and (2) and in particular satisfy the property that the composition with $\Omega^p_{A/\mathbf{Z}} \to \Omega^p_{B/\mathbf{Z}}$ is multiplication by $\delta(NL_{B/A})$.

Lemma 1.2. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic and assume given maps $c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \to f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$ for $p \geq 0$ satisfying (1) and (2). Then there is at most one collection of maps $c^p_{Y'/X'} : \Omega^p_{Y'/\mathbf{Z}} \to (f')^*\Omega^p_{X'/\mathbf{Z}} \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'})$ for $p \geq 0$ satisfying (1) and (2) such that the diagrams

$$\begin{array}{ccccc} b^*\Omega^p_{Y/\mathbf{Z}} & \xrightarrow[b^*c^p_{Y/X}]{} & b^*(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})) & = & (f')^*a^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} b^*\det(NL_{Y/X})) \\ \downarrow & & & & \downarrow \\ \Omega^p_{Y'/\mathbf{Z}} & & \xrightarrow{c^p_{Y'/X'}} & & (f')^*\Omega^p_{X'/\mathbf{Z}} \otimes_{\mathcal{O}_{Y'}} \det(NL_{Y'/X'}) \end{array}$$

commute for all $p \geq 0$. Here the vertical arrows use the maps $b^*\Omega^p_{Y/\mathbf{Z}} \to \Omega^p_{Y'/\mathbf{Z}}$ and $a^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{X'/\mathbf{Z}}$ of Section section de rham complex (Tag 07HX) as well as the identification $b^*\det(NL_{Y/X}) = \det(NL_{Y'/X'})$ of Discriminants, Section section tate map (Tag 0FKB).

证明. The map

$$(f')^*\Omega_{X'/\mathbf{Z}} \oplus b^*\Omega_{Y/\mathbf{Z}} \longrightarrow \Omega_{Y'/\mathbf{Z}}$$

is surjective because Y' is an open subscheme of the fibre product; the corresponding algebra statement is that $\Omega_{B\otimes_A C/\mathbf{Z}}$ is generated as a module by the images of $\Omega_{B/\mathbf{Z}}$ and $\Omega_{C/\mathbf{Z}}$. Thus for all p the map

$$\bigoplus\nolimits_{i=0,\ldots,p} (f')^*\Omega^i_{X'/\mathbf{Z}} \otimes b^*\Omega^{p-i}_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{Y'/\mathbf{Z}}$$

is surjective. Conditions (1) and (2) combined with the commutativity of the diagram in the lemma, implies that the map $c^p_{Y'/X'}$ is determined (but existence is not immediate). □

Lemma 1.3. *Consider a commutative diagram*

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

of schemes which induces an isomorphism of Y' with an open subscheme of $X' \times_X Y$. Assume f is locally quasi-finite and syntomic. Then for every $y' \in Y'$ we can find opens $V' \subset Y'$, $V \subset Y$, $U' \subset X$, $U \subset X$ with $y' \in V'$, with $f'(V') \subset U'$, $b(V') \subset V$, $a(U') \subset U$, $f(V) \subset U$, and such that for $p \geq 0$ there are maps $c^p_{V/U}$ and $c^p_{V'/U'}$ satisfying (1) and (2) which are compatible with the diagram

$$\begin{array}{ccc} V' & \xrightarrow[b]{} & V \\ \downarrow & & \downarrow \\ U' & \longrightarrow & U \end{array}$$

in the sense explained in Lemma 1.2.

证明. It is clear that we may replace Y' by $X' \times_X Y$ in order to prove this. Pick affine opens $V \subset Y$ and $U \subset X$ with V containing the image of y' as in Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE) part (5). Pick an affine open $U' \subset X'$ containing the image of y' and mapping into U. After replacing X, Y, X', Y' by $U, V, U', U' \times_U V$ we may assume we are in the situation described in the next paragraph.

Assume that $X' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, and $Y' = \mathrm{Spec}(A' \otimes_A B)$ with $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ a relative global complete intersection over A. Then also $B' = A'[x_1, \ldots, x_n]/(f'_1, \ldots, f'_n)$ is a relative global complete intersection over A' where f'_i is the image of f_i in $A'[x_1, \ldots, x_n]$, see Algebra, Lemma lemma base change relative global complete intersection (Tag 00SS). The construction in Remark 1.1 provides us with the maps $c^p_{V/U}$ and $c^p_{V'/U'}$. These maps are compatible by Lemma lemma funny map functorial (Tag 0H9F) and the compatibility of the presentations of B and B' over A and A'. Some details omitted. □

Example 1.4. Let $n \geq 1$ and $d \geq 1$ be integers. Let T be the set of multi-indices $E = (e_1, \ldots, e_n)$ with $e_i \geq 0$ and $\sum e_i \leq d$. Consider the ring

$$A = \mathbf{Z}[a_{i,E}; 1 \leq i \leq n, E \in T]$$

In $A[x_1, \ldots, x_n]$ consider the elements $f_i = \sum_{E \in T} a_{i,E} x^E$ where $x^E = x_1^{e_1} \ldots x_n^{e_n}$ as is customary. Consider the A-algebra

$$B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$$

Denote $X_{n,d} = \mathrm{Spec}(A)$ and let $Y_{n,d} \subset \mathrm{Spec}(B)$ be the maximal open subscheme such that the restriction of the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A) = X_{n,d}$ is quasi-finite, see Algebra, Lemma lemma quasi finite open (Tag 00QA).

Definition 1.5. Let $R \to S$ be a ring map. We say that $R \to S$ is a *relative global complete intersection* if there exists a presentation $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ and every nonempty fibre of $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ has dimension $n - c$. We will say "let $S = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ be a relative global complete intersection" to indicate this situation.

Primary problem: Tag 0FLA

Lemma 1.6. *There exists a unique rule that to every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_Y$-module maps*

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for $p \geq 0$ satisfying (1), (2), and (3). In particular, the composition of $c^p_{Y/X}$ with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$.

证明. This proof is very similar to the proof of Discriminants, Proposition proposition tate map (Tag 0FKD) and we suggest the reader look at that proof first.

Let us reformulate the statement. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $f : Y \to X$ of schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The lemma means that for every object Y/X of $\mathcal{C}$ we have maps $c^p_{Y/X}$, $p \geq 0$ with properties (1) and (2), and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have that $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 1.2.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine open $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ such that there exists some maps

$$\Omega^p_{Y/\mathbf{Z}}|_V \longrightarrow \left(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})\right)|_V$$

with properties (1) and (2). This follows from picking affine opens as in Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE) part (5) and then using the construction in Remark 1.1.

Note that the étale locus of f is exactly the set of points where $\delta(NL_{Y/X})$ does not vanish, see discussion in Discriminants, Section section tate map (Tag 0FKB). In particular $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ becomes an isomorphism after inverting $\delta(NL_{Y/X})$. We conclude that if $\Omega^p_{X/\mathbf{Z}}$ is finite locally free and the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero, then the local maps constructed in the previous paragraph are unique and automatically glue!

Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that

(1) $\Omega_{X/\mathbf{Z}}$ is locally free, and
(2) the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero.

By the remarks in the previous paragraph, we see that for any object Y/X of $\mathcal{C}_{nice}$ we have unique maps $c^p_{Y/X}$, $p \geq 0$ satisfying conditions (1) and (2). If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}_{nice}$, then the maps $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma 1.2: namely, locally there do exist compatible maps by Lemma 1.3 and the uniqueness just mentioned shows these maps agree with $c^p_{Y/X}$ and $c^p_{Y'/X'}$. In other words, we have solved the problem on the full subcategory $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c^p_{Y/X}$ the solution we've just found.

In fact, more generally, suppose we have a morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ such that Y/X is an object of $\mathcal{C}_{nice}$. The same argument, but this time using the uniqueness in Lemma 1.2, tells us there are maps $c^p_{Y'/X'}$, $p \geq 0$ satisfying conditions (1) and (2) compatible with the already constructed maps $c^p_{Y/X}$. Let us call this the base change $(b/a)^* c^p_{Y/X}$.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** The two base changes $(b_i/a_i)^* c^p_{Y_i/X_i}$, $i = 1, 2$ are equal. We will first show that the claim implies the lemma and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Discriminants, Example 1.4. Then $Y_{n,d}$ and $Y_{n,d}$ are irreducible schemes of finite type and smooth over $\mathbf{Z}$. Namely, $X_{n,d}$ is a spectrum of a polynomial ring over $\mathbf{Z}$ and $Y_{n,d}$ is an open subscheme of such. The

morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Discriminants, Lemma lemma universal quasi finite syntomic etale (Tag 0FK9). Thus $\delta(NL_{Y_{n,d}/X_{n,d}})$ is nonzero: for example we have the local description of $\delta(NL_{Y/X})$ in Discriminants, Remark remark local description delta (Tag 0FKC) and we have the local description of étale morphisms in Morphisms, Lemma lemma etale at point (Tag 02GU) part (8). Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Discriminants, Lemma lemma locally comes from universal (Tag 0FKA) we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Then we have the base change $c^p_{V/U} = (b/a)^* c^p_{Y_{n,d}/X_{n,d}}$. The claim guarantees these locally constructed maps $c^p_{V/U}$ glue! Thus we get a well defined global maps $c^p_{Y/X}$ with properties (1) and (2). Finally, let $b/a : Y'/X' \to Y/X$ be an arbitray morphism of $\mathcal{C}$. We have the maps $c^p_{Y'/X'}$, $p \geq 0$ and $c^p_{Y/X}$, $p \geq 0$ constructed in this paragraph. To check they are compatible as in Lemma 1.2 we may work locally on Y'/X' and Y/X. Thus we may assume that there exists a morphism $Y/X \to Y_{n,d}/X_{n,d}$. By construction, both the family $c^p_{Y'/X'}$ and the family $c^p_{X/Y}$ are compatible with the map towards $Y_{n,d}/X_{n,d}$. It follows easily from this (and the uniqueness in Lemma 1.2) that $c^p_{Y'/X'}$ is compatible with $c^p_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma lemma standard open two affines (Tag 0H9B). Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A'_1 = A_1[x_\alpha] \to A \quad \text{and} \quad A'_2 = A_2[y_\beta] \to A$$

of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A'_i \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

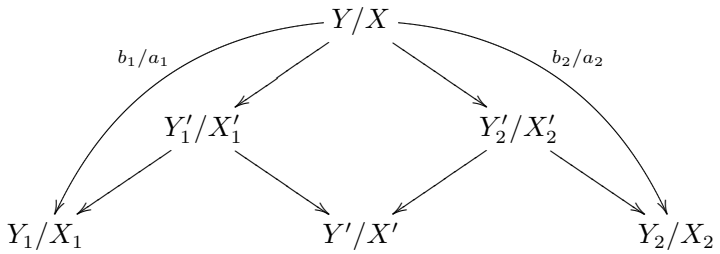
$$\begin{array}{ccccc} (A'_1 \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A'_2 \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A'_1 & \longrightarrow & A & \longleftarrow & A'_2 \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

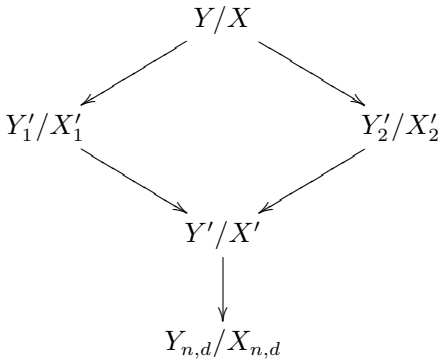
$$A' = A'_1 \times_A A'_2 \longrightarrow B' = (A'_1 \otimes_{A_1} B_1)_{h_1} \times_B (A'_2 \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma lemma module over fibre product (Tag 07RU) we have $A'_1 \otimes_{A'} B' = (A'_1 \otimes_{A_1} B_1)_{h_1}$ and $A'_2 \otimes_{A'} B' = (A'_2 \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' =$

$\mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE)). By More on Algebra, Lemma lemma properties algebras over fibre product (Tag 08KQ) the ring map $A' \to B'$ is flat and of finite presentation. Thus by Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE) part (3) we conclude that Y'/X' is an object of $\mathcal{C}$. Consider now the commutative diagram



where Y_i'/X_i' corresponds to $A_i' \to (A_i' \otimes_{A_i} B_i)_{h_i}$. Note that Y_i'/X_i' is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an infinite dimensional affine space with Y_i/X_i; small detail omitted. In particular, the pullback of $c^p_{Y_i/X_i}$ via (b_i/a_i) is the same as the pullback of $c^p_{Y_i'/X_i'}$ via $Y/X \to Y_i'/X_i'$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$, but this is almost never the case. Namely, then pulling back $c^p_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X_1', Y_1', X_2', Y_2', X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c^p_{Y_{n,d}/X_{n,d}}$. This finishes the proof. $\square$